

Hardware Attacks on ML

1. AttackGNN: Red-Teaming GNNs in Hardware Security Using Reinforcement Learning
2. INSIGHT: Attacking Industry-Adopted Learning Resilient Logic Locking Techniques Using Explainable Graph Neural Network

Federated Learning Security

3. FAMOS: Robust Privacy-Preserving Authentication on Payment Apps via Federated Multi-Modal Contrastive Learning
4. Efficient Privacy Auditing in Federated Learning
5. Defending Against Data Reconstruction Attacks in Federated Learning: An Information Theory Approach
6. Lotto: Secure Participant Selection against Adversarial Servers in Federated Learning
7. Lurking in the shadows: Unveiling Stealthy Backdoor Attacks against Personalized Federated Learning
8. ACE: A Model Poisoning Attack on Contribution Evaluation Methods in Federated Learning
9. BackdoorIndicator: Leveraging OOD Data for Proactive Backdoor Detection in Federated Learning

Fault Injection & Robustness

10. DNN-GP: Diagnosing and Mitigating Model's Faults Using Latent Concepts
11. Yes, One-Bit-Flip Matters! Universal DNN Model Inference Depletion with Runtime Code Fault Injection
12. Tossing in the Dark: Practical Bit-Flipping on Gray-box Deep Neural Networks for Runtime Trojan Injection
13. Forget and Rewire: Enhancing the Resilience of Transformer-based Models against Bit-Flip Attacks

Deepfake & Synthesis

14. Double Face: Leveraging User Intelligence to Characterize and Recognize AI-synthesized Faces
15. SoK: The Good, The Bad, and The Unbalanced: Measuring Structural Limitations of Deepfake Media Datasets
16. Can I Hear Your Face? Pervasive Attack on Voice Authentication Systems with a Single Face Image
17. dp-promise: Differentially Private Diffusion Probabilistic Models for Image Synthesis

ML for Security (Detection/Classification)

- 18. Automated Large-Scale Analysis of Cookie Notice Compliance
- 19. Detecting and Mitigating Sampling Bias in Cybersecurity with Unlabeled Data
- 20. Code is not Natural Language: Unlock the Power of Semantics-Oriented Graph Representation for Binary Code Similarity Detection
- 21. VulSim: Leveraging Similarity of Multi-Dimensional Neighbor Embeddings for Vulnerability Detection

LLM Attacks & Defenses

- 22. An LLM-Assisted Easy-to-Trigger Backdoor Attack on Code Completion Models
- 23. REMARK-LLM: A Robust and Efficient Watermarking Framework for Generative Large Language Models
- 24. Formalizing and Benchmarking Prompt Injection Attacks and Defenses
- 25. Instruction Backdoor Attacks Against Customized LLMs
- 26. LLM-Fuzzer: Scaling Assessment of Large Language Model Jailbreaks
- 27. Don't Listen To Me: Understanding and Exploring Jailbreak Prompts of Large Language Models
- 28. Malla: Demystifying Real-world Large Language Model Integrated Malicious Services
- 29. Making Them Ask and Answer: Jailbreaking Large Language Models in Few Queries via Disguise and Reconstruction
- 30. Moderating Illicit Online Image Promotion for Unsafe User Generated Content Games Using Large Vision-Language Models
- 31. Deciphering Textual Authenticity: A Generalized Strategy through the Lens of Large Language Semantics for Detecting Human vs. Machine-Generated Text
- 32. Prompt Stealing Attacks Against Text-to-Image Generation Models
- 33. Quantifying Privacy Risks of Prompts in Visual Prompt Learning

Secure ML Inference

- 34. AutoFHE: Automated Adaption of CNNs for Efficient Evaluation over FHE
- 35. Fast and Private Inference of Deep Neural Networks by Co-designing Activation Functions
- 36. OblivGNN: Oblivious Inference on Transductive and Inductive Graph Neural Network
- 37. MD-ML: Super Fast Privacy-Preserving Machine Learning for Malicious Security with a Dishonest Majority
- 38. Accelerating Secure Collaborative Machine Learning with Protocol-Aware RDMA

Privacy Inference Attacks

- 39. A Linear Reconstruction Approach for Attribute Inference Attacks against Synthetic Data

- 40. Did the Neurons Read your Book? Document-level Membership Inference for Large Language Models
- 41. MIST: Defending Against Membership Inference Attacks Through Membership-Invariant Subspace Training
- 42. Inf2Guard: An Information-Theoretic Framework for Learning Privacy-Preserving Representations against Inference Attacks
- 43. Property Existence Inference against Generative Models
- 44. How Does a Deep Learning Model Architecture Impact Its Privacy? A Comprehensive Study of Privacy Attacks on CNNs and Transformers
- 45. Reconstructing training data from document understanding models
- 46. Privacy Side Channels in Machine Learning Systems
- 47. FaceObfuscator: Defending Deep Learning-based Privacy Attacks with Gradient Descent-resistant Features in Face Recognition

Backdoor Attacks & Defenses

- 48. Neural Network Semantic Backdoor Detection and Mitigation: A Causality-Based Approach
- 49. On the Difficulty of Defending Contrastive Learning against Backdoor Attacks
- 50. Mudjacking: Patching Backdoor Vulnerabilities in Foundation Models
- 51. Xplain: Analyzing Invisible Correlations in Model Explanation
- 52. Verify your Labels! Trustworthy Predictions and Datasets via Confidence Scores
- 53. UBA-Inf: Unlearning Activated Backdoor Attack with Influence-Driven Camouflage
- 54. Devil in the Room: Triggering Audio Backdoors in the Physical World

Adversarial Evasion Attacks

- 55. More Simplicity for Trainers, More Opportunity for Attackers: Black-Box Attacks on Speaker Recognition Systems by Inferring Feature Extractor
- 56. Transferability of White-box Perturbations: Query-Efficient Adversarial Attacks against Commercial DNN Services
- 57. Adversarial Illusions in Multi-Modal Embeddings
- 58. It Doesn't Look Like Anything to Me: Using Diffusion Model to Subvert Visual Phishing Detectors
- 59. Invisibility Cloak: Proactive Defense Against Visual Game Cheating

Adversarial Defenses & Robustness

- 60. Correction-based Defense Against Adversarial Video Attacks via Discretization-Enhanced Video Compressive Sensing
- 61. Rethinking the Invisible Protection against Unauthorized Image Usage in Stable Diffusion

- 62. Splitting the Difference on Adversarial Training
- 63. Machine Learning needs Better Randomness Standards: Randomised Smoothing and PRNG-based attacks
- 64. PatchCure: Improving Certifiable Robustness, Model Utility, and Computation Efficiency of Adversarial Patch Defenses
- 65. FraudWhistler: A Resilient, Robust and Plug-and-play Adversarial Example Detection Method for Speaker Recognition

Model Extraction & Watermarking

- 66. SoK: All You Need to Know About On-Device ML Model Extraction - The Gap Between Research and Practice
- 67. Unveiling the Secrets without Data: Can Graph Neural Networks Be Exploited through Data-Free Model Extraction Attacks?
- 68. ClearStamp: A Human-Visible and Robust Model-Ownership Proof based on Transposed Model Training
- 69. DeepEclipse: How to Break White-Box DNN-Watermarking Schemes
- 70. moDelGuard: Information-Theoretic Defense Against Model Extraction Attacks

ML Security Analysis

- 71. SECUritynet: Assessing Machine Learning Vulnerabilities on Public Models
- 72. Hijacking Attacks against Neural Network by Analyzing Training Data
- 73. False Claims against Model Ownership Resolution
- 74. Landscape More Secure Than Portrait? Zooming Into the Directionality of Digital Images With Security Implications
- 75. Information Flow Control in Machine Learning through Modular Model Architecture

ML-Based Vulnerability Detection

- 76. Racing on the Negative Force: Efficient Vulnerability Root-Cause Analysis through Reinforcement Learning on Counterexamples
- 77. Uncovering the Limits of Machine Learning for Automatic Vulnerability Detection
- 78. Improving ML-based Binary Function Similarity Detection by Assessing and Deprioritizing Control Flow Graph Features
- 79. tyGr: Type Inference on Stripped Binaries using Graph Neural Networks
- 80. EaTVul: ChatGPT-based Evasion Attack Against Software Vulnerability Detection
- 81. FVD-DPM: Fine-grained Vulnerability Detection via Conditional Diffusion Probabilistic Models
- 82. A Wolf in Sheep's Clothing: Practical Black-box Adversarial Attacks for Evading Learning-based Windows Malware Detection in the Wild

Physical Adversarial Attacks

- 83. π -Jack: Physical-World Adversarial Attack on Monocular Depth Estimation with Perspective Hijacking
- 84. AE-Morpher: Improve Physical Robustness of Adversarial Objects against LiDAR-based Detectors via Object Reconstruction
- 85. A First Physical-World Trajectory Prediction Attack via LiDAR-induced Deceptions in Autonomous Driving
- 86. On Data Fabrication in Collaborative Vehicular Perception: Attacks and Countermeasures
- 87. voGUEs: Validation of Object Guises using Estimated Components
- 88. Adversary is on the Road: Attacks on Visual SLAM using Unnoticeable Adversarial Patch